

Part A

1. Consider an exchange economy with two individuals, A and B, and two goods, x and y . Individual A has an endowment of $\omega_A = (5, 1)$ and individual B has an endowment of $\omega_B = (1, 5)$. The utility functions of the individuals are, respectively:

$$u_A(x_A, y_A) = x_A^2 y_A^3,$$

$$u_B(x_B, y_B) = \min\{2x_B, 3y_B\}.$$

(a) Normalise the price of good x to 1, and denote by p the price of good y .

- i. Verify whether the aggregate excess demand for good x at price p is equal to $\frac{10+2p}{5} + \frac{3+15p}{3+2p} - 6$.
- ii. How does the aggregate excess demand vary with p ? How is this related to the existence of a competitive equilibrium?

(b) Consider now price $p = 1$.

- i. What is the aggregate excess demand for good x at this price? Does this price correspond to an equilibrium? Is the corresponding equilibrium allocation Pareto efficient?
- ii. Illustrate with an Edgeworth box, including the endowments, the budget constraints and the relevant indifference curves of both individuals.

[25%]

2. Two players, *Row* and *Col*, are playing the following game. Each player has the same available actions X and Y . The payoffs are symmetric, and represented in the matrix below, with the left payoff in each cell being that of *Row* and the right payoff in each cell being that of *Col*.

		<i>Col</i>	
		X	Y
<i>Row</i>	X	a, a	b, c
	Y	c, b	d, d

(a) Consider first that the players move simultaneously.

- i. If $a > c$ and $b > d$, what are the Nash equilibria? Is this a Prisoner's Dilemma?
- ii. If $a > c$ and $b < d$, what are the Nash equilibria, both in pure and mixed strategies?

(b) Consider now that *Row* moves first, and *Col* knows the strategy that *Row* has chosen.

- i. Explain the equilibrium concept that best applies to this type of situation.
- ii. If $b > d > c > a$, what is the equilibrium?

[25%]

3. Lottery X pays prizes x_1 and x_2 with equal probability, while lottery Y pays prizes y_1 and y_2 with equal probability.

(a) Assume that individuals like money. Explain whether the following statements are true or not:

- i. If $y_1 < x_1 < x_2 < y_2$, any risk averse expected utility maximiser prefers X to Y .
- ii. If $y_1 < x_1 < x_2 < y_2$ and the expected value of X is above the expected value of Y , any risk averse expected utility maximiser prefers X to Y .

(b) An individual, who again likes money, prefers lottery X to lottery Y . You offer this individual two modified lotteries: Compound lottery X' consists of a coin flip with the payment of lottery X when heads, and zero when tails; compound lottery Y' consists of a coin flip with the payment of zero when heads and the payment of lottery Y when tails. Explain whether the following statements are true or not:

- i. If the individual is an expected utility maximiser, X' will be preferred over Y' .
- ii. If the individual maximises a utility function, X' will be preferred over Y' .

[25%]

4. All households have the same wealth w , are risk-averse, and face a potential loss $K < w$. Half of the households are low-risk and have a probability $\frac{1}{4}$ of incurring the loss. Half of the households are high-risk, with a probability of $\frac{1}{2}$ of incurring the loss.

Insurance companies operate in a competitive market, are risk-neutral, and have no administrative costs.

- (a) Suppose that types are observable by insurers. Regulators 1 and 2 propose that insurance companies set different prices for different households, p_L and p_H per pound of cover. While Regulator 1 proposes that all households must fully insure, Regulator 2 proposes that they can partially insure (for no more than their total loss K). Explain why, in equilibrium, households may find both regulations equivalent. What are the equilibrium prices p_L, p_H ?
- (b) Regulators 3 and 4 propose that insurance companies must set the same price for all households, p per pound of cover.
 - i. Regulator 3 proposes that all households must fully insure. What is the equilibrium price p ?
 - ii. Regulator 4 proposes allowing households to partially insure (for no more than the total loss K). Denote by $q(p)$ the low-risk household's demand for insurance and assume that $q'(p) < 0$ holds for prices in $\frac{1}{2} \geq p \geq \frac{1}{4}$. Verify whether there is a unique equilibrium price, which may be written as a weighted average of $\frac{1}{2}$ and $\frac{1}{4}$, at which insurance companies break even in expectation.

[25 %]

Part B

5. The senior tutor in an Oxford college wants to rank all the students in a group, in order to allocate a prize. In this ranking she wants to aggregate the opinions that different tutors have over these students. If you had to design a social choice rule, or some other mechanism, to help her with this task, which features should it have and why?

6. “Positive externalities have, by their nature, a positive impact on society and can therefore be disregarded by policy makers.” Discuss.

7. Suppose that two firms in a duopoly currently act non-cooperatively. They propose a merger. Outline how oligopoly theory can help the competition authority to determine the conditions under which the proposed merger should be allowed. How would your analysis of the merger of these two firms change if there were initially more than two firms in the industry (while it is still two firms who plan to merge)?

8. Suppose that you are a financial advisor, that all your clients are expected utility maximisers and that you want to give them individualised advice. What exact information, about their utility function over money and other relevant factors, is fundamental in order to understand their risk attitudes? Why is the Arrow-Pratt theorem relevant in this context? How would you design a survey using lotteries to obtain the desired information from client responses?

9. Consider a risk-neutral principal and an agent. Both the principal and the agent observe the realised profit level (which can be low or high), but only the agent knows the effort that they put in (which can be low or high). Evaluate the truth of the following statements, explaining your reasoning.

1. There is a positive agency cost if and only if the principal chooses to induce high effort and the agent is risk-averse.
2. With risk-averse agents, the more risk-averse the agent, the higher the agency cost.
3. With risk-averse agents, the larger the difference between the high and low profit levels, the larger the difference between wages in the two states of the world when the principal wants to induce high effort.

10. College 1 and College 2 want to establish a joint library, with books available to members of both colleges. The preferences of both colleges are the same, and are represented by the utility function $u_i(B, x_i) = \frac{1}{3} \ln B + \frac{2}{3} \ln x_i$, where $B = b_1 + b_2$ is the total number of books in the joint library and x_i is the money that college i has for other uses. Each college has an initial endowment of £600 of money and no books. The price of books is $p_b = 1$ and the price of money is $p_x = 1$.

(a) Suppose that each college decides, independently of the other, how many books to purchase and put in the joint library.

- i. Show that, if college 2 were to buy b_2 books, where $b_2 < 300$, college 1 would buy $b_1 = 200 - \frac{2}{3}b_2$ books.
- ii. How many books in total would there be in the library when each college chooses independently? 30

(b) Consider now the optimal joint provision of books.

- i. Write down the condition that an optimal provision of books must satisfy.
- ii. What is the optimal number of books and why is it different from the number of books purchased when each college decides independently of the other? 30

(c) Suppose now that the university wants to help colleges to implement the socially optimal number of books.

- i. Describe formally how the Lindahl pricing mechanism would work in this situation. (You may assume that in the social optimum each college pays for half the optimal number of books.)
- ii. Describe formally why the notion of strategy-proofness is relevant in this type of problem.
- iii. The university decides to let colleges act independently, and pays for a subsidy for books so the new price is below 1. What price will ensure that colleges buy the optimal number of books? [40%]

Solution to 1.

- (a) The demand of the first individual can be obtained from the standard optimisation program, but the student could easily derive it from the properties of Cobb-Douglas preferences. That is, we know that a fraction $\frac{2}{5}$ of the total wealth will be allocated to good x . Hence,

$$x_A^* = \frac{10 + 2p}{5}.$$

The demand of the second individual can be derived by simply arguing that, with Leontieff preferences, it must be $2x_B = 3y_B$. Hence, using the budget constraint, we must have $x_B + p\frac{2}{3}x_B = 1 + 5p$, and hence:

$$x_B^* = \frac{3 + 15p}{3 + 2p}.$$

Aggregate excess demand is the sum of these two demands minus the total endowment, which is 6, hence the result.

We can simply derivate aggregate excess demand with respect to p , to obtain:

$$\frac{2}{5} + \frac{15(3 + 2p) - (3 + 15p)2}{(3 + 2p)^2} = \frac{2}{5} + \frac{39}{(3 + 2p)^2} > 0.$$

As it should be expected, an increase in the price of good y makes the excess demand of good x to increase. Now, notice that when p is very small, the excess demand of x is negative (approaches -3), while if p is very large, it is positive (approaches ∞). The student can thus be very formal here and suggest that the function is continuous and must cross 0, which will correspond to an equilibrium (via Walras' law). Obviously, partial arguments are to be expected. In any case, they are helped in the second part to derive similar conclusions.

- (b) For price $p = 1$, aggregate demand is 6 and given that the total endowment of good x is 6, aggregate excess demand is zero. The situation must correspond to an equilibrium because the other market must also be in equilibrium (due to Walras' Law). The equilibrium allocation can be immediately obtained from the existing information, corresponding to $\{(\frac{12}{5}, \frac{18}{5}), (\frac{18}{5}, \frac{12}{5})\}$ and is efficient from the first welfare theorem (any argument equating MRS must be scrutinised because individual B has kinky indifference curves). The graph must identify this allocation, the endowment $(5, 1)$, the budget set with slope 1 passing through the endowment and the equilibrium, at least one convex indifference curve for A with the tangency condition at the equilibrium point, and at least one Leontieff indifference curve for B with kink at the equilibrium.

Solution to 2.

- (a) In the first situation, $\{X, X\}$ is a dominant strategy equilibrium. There are no other equilibria. Notice that it is a Prisoner's Dilemma only if $d > a$, in which case both would be better off with the non-equilibrium outcomes $\{Y, Y\}$ than with the equilibrium ones. However, if $a \geq d$, the equilibrium is Pareto efficient and the situation is not a Prisoner's Dilemma.

The second situation corresponds to a coordination game. There are two pure strategy Nash equilibria that can be easily identified and correspond to $\{X, X\}$ and $\{Y, Y\}$. There is also a mixed strategy equilibrium. Suppose *Col* plays X with probability $1 - p$ and Y with probability p . *Row* is indifferent between the pure strategies if and only if

$$(1 - p)a + pb = (1 - p)c + pd$$

so

$$p = \frac{a - c}{a - c + d - b}$$

which is positive and below 1. If p is different to that value, one strategy is strictly better and no randomisation is possible. The game is symmetric so the same for the other player. Hence, the mixed strategy equilibrium has each player playing Y with that probability and X with the complementary probability

$$1 - p = \frac{d - b}{a - c + d - b}.$$

- (b) The appropriate concept is sub-game perfect (Nash) equilibrium, i.e., Nash equilibrium in the game and in every sub-game. In a finite game, this can be found using backwards induction. Some theoretical discussion is to be expected.

In this simple game, the sub-game perfect equilibrium corresponds to $\{X; Y \text{ if } X, X \text{ if } Y\}$. Note that *Row*'s strategy is just the action X , while *Col*'s equilibrium strategy is a complete contingency plan, specifying what would happen in each possible situation *Col* might be in. It is likely that many will say that *Col*'s strategy is just Y , which isn't the full statement.

Solution to 3.

- (a) The first statement is false. As part of the discussion, the student can mention that payments x_1 and x_2 may be very close to y_1 , while y_2 could be very large. This allows to consider cases in which expected value is increased substantially, at almost no risk. A formal proof can be done by showing a counter-example. For instance, consider CRRA preferences with $m^{\frac{1}{2}}$ and payments $1 < 4 < 9 < 100$, bringing expected utilities 2.5 and 5.5.

The second part is true. As part of the discussion, notice that lottery X has now a larger expected value with less variability. A formal proof can use a lottery Z where you simply subtract some amount of money from x_1 and x_2 , to equalise the expected value of the lotteries, while keeping these values above y_1 . Then, Z SOSDs Y and the risk averse individual will prefer Z to Y . X FOSDs Z and every expected utility maximiser should prefer X to Z . Transitivity concludes the argument.

- (b) The first statement is true. As part of the discussion, notice that EU maximisers only care about the simple lotteries induced (and hence, whether something is paid with heads or, alternatively, with tails, should be irrelevant) and the rest of the idea is merely the independence property, that students should mention. For the proof, the student can simply use the independence property or, alternatively, the EU representation and claim that, for an expected utility maximiser, we can write $EU(X) \geq EU(Y)$ if and only if $EU(X') = \frac{1}{2}EU(X) + \frac{1}{2}EU(0) \geq \frac{1}{2}EU(Y) + \frac{1}{2}EU(0) = EU(Y')$.

The second statement is false. As part of the discussion, notice that maximising utility is not enough because the utility may be in flagrant conflict with the compound lotteries principle, with independence or with continuity. The students may suggest any example violating any of these principles, but notice that no complicated example is truly needed. There are only four lotteries, so utility function $u(X) = 4, u(Y) = 3, u(Y') = 2, U(X') = 1$ is enough.

Solution to 4.

- (a) For a typical firm, the expected profits from supplying the low-risk type with partial insurance cover of q , at a price p_L , is $p_L q - \frac{1}{4}q$. With competition and risk-neutrality, expected profit will be zero so $p_L = \frac{1}{4}$. Similarly, $p_H = \frac{1}{2}$. Now, since insurance is actuarially fair, every risk averse household would like to fully insure. This can be shown by writing expected utility, for the low-risk type say, as

$$\frac{3}{4}u(w - \frac{1}{4}q) + \frac{1}{4}u(w - \frac{1}{4}q - K + q).$$

The first-order condition for the choice of q leads to $u'(w - \frac{1}{4}q) = u'(w - K + (1 - \frac{1}{4})q)$ which can only be true when $q = K$ (The second-order condition holds, by concavity; another way to show this is to note that expected wealth when $p = \frac{1}{4}$ is $w - \frac{1}{4}K$, independently of the value of q and hence, SOSDs makes full insurance to dominate any other level of insurance). The same logic applies to the high-risk group. Since individuals will insure fully, there is no difference between both regulations.

- (b) With full insurance, each type buys insurance of K . The expected profit of an insurance firm for a contract for one household is $pK - \frac{1}{2}\frac{1}{4}K - \frac{1}{2}\frac{1}{2}K$, which equals zero when $p = \frac{3}{8}$. With partial insurance, notice that if $p = \frac{1}{4}$, then the low-risk types fully insure, as do the high-risk types, who get a very good deal. The firms make losses on the high-risk types and this cannot be an equilibrium because firms would prefer to leave the market. If $p = \frac{1}{2}$, the high-risk types fully insure and the firms make zero expected profits on them. The low-risk types buy less than full cover (because $q'(p) < 0$ and they buy full cover when $p = \frac{1}{4}$; naturally, students may use risk aversion and some first order condition to discuss this property of demand, but it is given and not needed), and the firms make profits on these contracts. This cannot be an equilibrium because it would attract entry. The equilibrium price satisfies $\frac{1}{4} < p < \frac{1}{2}$. The typical firm will have half of its customers being low risks, who buy $q(p) < K$, and half being high risks, who insure fully. Expected profits are

$$\frac{1}{2}(p - \frac{1}{4})q(p) + \frac{1}{2}(p - \frac{1}{2})K$$

which are zero when the equilibrium price is

$$p = \frac{\frac{1}{4}q(p) + \frac{1}{2}K}{q(p) + K}.$$

Sketch solution to essay 5

Students should show their knowledge of the topic by appropriately discussing properties, rules and information that fits reasonably the problem.

Given the wording, social choice rules are expected to be discussed in length. Notice that mechanisms that simply choose the best option may be partially discussed here, because although a ranking must be done, only one prize is to be given. As students will surely mention them, emphasis on why would Arrow properties are relevant is key. In particular, notice that universal domain is important here because single-peaked preferences are not to be expected (but very smart examples could be constructed, i.e., the group corresponds to students taking two disciplines, all students are equally smart but simply combine disciplines to a different extent and tutors just like those who are closer to them). That is, horizontal heterogeneity and single-peakedness is possible, but the student should properly present it. References to non-manipulability may be also relevant.

Some references to cardinal considerations seem very relevant here. Tutors surely have cardinal information on the quality of students, based upon collections, exercises, etc. Then, one should expect a serious discussion of cardinal considerations, with emphasis on efficiency and equality. Importantly, the *mechanism* would then require this information to be cardinal and comparable, leading to implementation difficulties (manipulability may be an issue). Many other responses and analysis are possible. Within an open-book exam, it is particularly important to write an articulate and organised essay that places particular emphasis in the exact questions and details of the exam statement.

Sketch solution to essay 6

The statement is clearly wrong and students are expected to make some claim based upon the market failures when externalities exist, either positive or negative.

The main approach to the essay will likely present some formal definition of positive externalities, with some example (vaccination may be a possible route in the current context). Importantly, the essay should not be written from the point of a negative externality. A discussion on the absence of market for this type of situation is relevant. The first task would probably be to theoretically understand how a decentralised economy cannot produce a Pareto efficient allocation in this type of situation. Actors not internalising the positive effects of their actions in other actors are important, because if they do not consider them, the social marginal benefit or return of their actions will not be considered.

Then, the standard government interventions will be presented. Creation of bargaining opportunities recovering efficiency via the Coase theorem, or Pigouvian subsidies are to be expected. Quota discussions should be related to mandatory actions by parties (say, mandatory vaccination), etc. Special attention should be paid to the fact that these are positive and not negative externalities, and the discussion should just consider them (beyond some initial, general, discussion of the role of externalities maybe).

Sketch solution to essay 7. Students may discuss the two standard objectives for competition authorities: social welfare (the sum of consumer surplus and profits), and consumer surplus itself. Answers for the three most standard duopoly models may be presented.¹ For the last part, also the effect of a merger on collusion is relevant. Mentioning at least one of the three models and discussing collusion in the context of n firms should be enough for a 2.1 answer.

1.1 Identical-products Bertrand. Consumer surplus falls if price increases, i.e., when the new monopoly price is above the old marginal cost.² If price falls, allow the merger. If a social welfare standard is used, Williamson trade-off: merger allowed if the gain in profits exceeds the loss of consumer surplus. Obviously, a less stringent requirement than that consumer surplus rises.

1.2 Cournot duopoly. Market power increase is smaller than with Bertrand as there is some initial market power. For a given reduction in marginal cost, more likely to allow the merger.³

1.3 Differentiated products Bertrand. Similar to Cournot, with market power increase smaller than with identical products. Can use the Upward Pricing Pressure formula from the lecture. A increases its price if B 's pre-merger margin $\times$ Diversion Ratio $>$ A 's Reduction in Marginal Cost.⁴ Upward Price Pressure does not look at the new equilibrium but instead assesses the local incentive to change price at the pre-merger prices and with the new marginal costs.

2. Generalization to $n > 2$ firms. Candidates likely to address whether a merger of two firms is likely to increase collusion. If n falls it is more likely that collusion is sustainable in a repeated game framework.⁵ With identical products and price-setting the merging firms have no market power if their cost reduction is small. The market price will be unchanged while the merged firm becomes the only producer but constrained by the other firms with the old cost level. Profits rise but consumer surplus is constant. In Cournot, the merging firms, if they restrict output, provide an advantage to the non-merging firms, who increase their output (and this may be such that the merger is no longer profitable). Still bad for consumers as overall output will fall.⁶ In differentiated-products Bertrand the sign of the Upward Pricing Pressure formula determines what other firms do: if the merged firm increases prices then so do the outsiders to the merger, and consumers will certainly lose from the merger.

¹1.1 Identical products Bertrand, with firms setting prices, where prices equal the common marginal cost; 1.2 Cournot, with firms choosing outputs, yielding an outcome with duopoly in between monopoly and the Bertrand outcome; 1.3 differentiated-products Bertrand, yielding equilibrium prices above marginal costs.

²In IO language, not required, this would be a non-drastic innovation.

³Those who have done the Economics of Industry option might note that total output will rise, and thus the price will fall, if the marginal cost falls sufficiently and that the initial Cournot first-order conditions can be used to determine how much the cost should fall. Not required here.

⁴Diversion ratio is the fraction of sales lost by A from a price increase that go to B , i.e., $D_{AB} = \frac{\partial Q_B / \partial p_A}{-\partial Q_A / \partial p_A}$.

⁵For discount factor δ , condition for sustainability of collusion, $\delta > \frac{n-1}{n}$, is more likely to hold as n falls.

⁶Those who have done the Economics of Industry option may state that, with Cournot and linear demand, if the merging firms' combined share is below that of the non-merging firms then a price-increasing merger will raise the combined "external" welfare of the rival firms and consumers together: this is not required for a very good answer as those who've done only Microeconomics won't have covered that.

Sketch solution to essay 6

Fundamentally, as all clients are expected utility maximisers, individualised advice is relevant only if there is variation among clients, and a discussion of variation should be done somewhere. A good student may surely point out that the investment part is not the only source of risk and that individualised advice would require to learn many things!

On the basic answer to the information question, it is evident that EU is fully captured by the utility u over money. An understanding of this function and the level of wealth of the individual will help to understand the risk attitude. Naturally, wealth may be irrelevant to some extent if preferences are Constant Absolute Risk Averse (CARA), and a discussion of this point is to be expected during the essay. For the understanding of u , concavity is the key element explaining risk attitudes. A graph explaining how concavity is the key concept would help. Study of the concavity of function u can be done throughout the Arrow-Pratt coefficients. A discussion of the normalisation of the second derivative is fundamental here, in connection to the fact that affine transformations of u represent the same expected utility preferences but have different second derivatives due to the scalar transformation. Correction of this scalar transformation throughout a ratio with the first derivative produces the Arrow-Pratt coefficients and a good understanding of the concavity of u . The notion of more concavity via concave transformations of u and/or the comparison of Arrow-Pratt coefficients is to be expected, because classifying clients in terms of risk groups/categories seems to be an obvious way to produce individualised (but not too costly individualised) advised. A classification by means of CARA or Constant Relative Risk Aversion (CRRA), that produce a unique summary of concavity with a single coefficient may be proposed.

Naturally, the Arrow-Pratt theorem is fundamental because it links risk attitudes in EU preferences with choice behavior. Thus, the financial advisor can learn preferences from simple choices in a survey, or claim that they will execute financial investments that the clients would choose themselves. The presentation of the theorem requires to discuss the equivalence of the comparison of Arrow-Pratt coefficients with the notion of more concavity and the comparison of Certainty Equivalents (with more risk averse individuals having always lower certainty equivalents). From there, students can also discuss the choice version of the theorem, i.e., if an individual rejects a lottery, a more risk averse individual will also do it, etc.

The survey question can thus be easily connected to the previous discussions, and allow students to show their understanding. The minimal response should claim that, because of the certainty equivalent relationship, you can ask clients whether they would accept/reject certain lotteries, and the more rejections you observe, the more risk aversion. The students have been presented the difference between CARA and CRRA, so they can discuss how the survey can present questions in terms of absolute lotteries or relative lotteries (i.e., win a fraction of your wealth, lose a fraction of your wealth). They can discuss how some lotteries are safer than others (via second-order stochastic dominance) and hence inform about risk aversion, etc.

Sketch solution to essay 9

The set-up is the standard two-state, two-effort level model.

1. The first statement is true. The agency cost is the increase in the expected wage, conditional on the effort level induced, above the expected wage when effort is observable and contractible. This higher expected wage is needed to cover the risk premium of the agent. To induce high effort the principal needs to pay a higher wage in the good state than in the bad state. The agent's incentive compatibility and participation constraints are just satisfied. If the agent were risk-neutral then there would be no risk premium and thus no agency cost.

If the principal chooses to induce low effort then the wage will be fixed, and equal to that which just ensures participation. There is no risk premium for the agent as the agent bears no risk. There is thus no increase in the expected wage above that with symmetric information.

2. The second statement is true. The agency cost arises because there is a risk premium, which is increasing in the agent's (absolute) risk aversion. This follows directly from the Arrow-Pratt theorem (lower certainty equivalent is the same than higher risk premium for any lottery, and the contingent payment of the principal is simply one such lottery; a remark on which the premium is, approximately $\frac{1}{2}A\sigma_w^2$, as a function of the variance of the wage may be done. A simpler and less general story is to note that if there is risk neutrality the risk premium is zero, while if there is risk aversion the premium is positive.

3. The third statement is false. The optimal wages in the two states are fully determined by the agent's incentive compatibility and participation constraints, and do not directly depend on the actual levels of profits in the two states. The profit levels will help to determine whether it is optimal to induce high or low effort, but there is no change in the wage structure that induces high effort when the profit levels change.

Solution to Exercise 10:

- (a) The utility function for 1, with $x_1 = 600 - b_1$, is

$$\frac{1}{3} \ln(b_1 + b_2) + \frac{2}{3} \ln(600 - b_1)$$

so the FOC for 1 is

$$\frac{1}{3} \frac{1}{b_1 + b_2} - \frac{2}{3} \frac{1}{600 - b_1} = 0$$

Solving for b_1 gives

$$b_1 = 200 - \frac{2}{3}b_2$$

as required.

Solving the two best-responses, or the two first-order conditions, gives $b_1 = b_2 = 120$ for the equilibrium provision per college with independent choices.

- (b) For the Samuelson condition, the sum of the MRSs should equal to $p = 1$. We can write $MU_b = \frac{1}{3} \frac{1}{b_1 + b_2}$ and $MU_{x_i} = \frac{2}{3} \frac{1}{600 - b_i}$ so

$$\frac{MU_b}{MU_{x_1}} + \frac{MU_b}{MU_{x_2}} = \frac{600 - b_1}{2(b_1 + b_2)} + \frac{600 - b_2}{2(b_1 + b_2)} = 1$$

so the total number of books bought should be

$$B = b_1 + b_2 = 400.$$

[It is natural to suppose that $b_1 = b_2 = 200$, though in fact the Samuelson condition is satisfied with any pair $\{b_1, b_2\}$ such that $b_1 + b_2 = 400$. If in addition we require the utilities to be equal and each college to pay for their own provision, i.e. 1 pays $pb_1 = b_1$ then it will follow that $b_1 = b_2$.]

- (c) For Lindahl, find the MRS at the optimum for each college and set p_i equal to this. With $b_1 = b_2$ the MRS at the optimum is $1/2$ of the relative price, i.e. $1/2$ for each college. So each college should choose the total amount of the public good, B , paying a price of $1/2$ per unit. The utility function is $\frac{1}{3} \ln B + \frac{2}{3} \ln(600 - 0.5B)$, and maximizing this with respect to B confirms that each college would choose $B = 400$. For strategy-proofness, if the utility functions are private information then Lindahl won't work: each is tempted to say that they value books less at the margin than is true, because this then reduces the individual price they pay. For the outside subsidy, this requires the Nash equilibrium to be solved for again with a general p . The best response is

$$b_1 = \frac{200}{p} - \frac{2}{3}b_2$$

and the Nash equilibrium is

$$b_1 = b_2 = \frac{120}{p}.$$

The subsidized price that ensures that $b_1 = b_2 = 200$ is found by

$$\frac{120}{p} = 200$$

so

$$p = \frac{120}{200} = \frac{6}{10} = 0.6.$$